Visualization and Dashboard

Formula 1 Race Intelligence – Part 3 of 4

1. Worksheets

The workbook contains 10 worksheets. Each has a caption that can be placed under the chart on the dashboard.

#	Worksheet	Chart	Fields	Caption
1	Team Championship Trajectory	Line	season, team, points, finish_position	Season-by-season championship points for each constructor, with Red Bull Racing and Mercedes highlighted against the selected team.
2	Tyre Strategy	Heat map / bars	starting_compound, circuit_type, is_wet_race, points	Starting tyre compound across circuit types, with a wet race filter, showing which compounds suit which tracks.
3	Talent Pools	Bar	nationality, points, Podium	Total points and podiums by driver nationality, showing the strongest talent pools for recruitment.
4	Driver Leaderboard	Bar	driver, season, points, pit_stops	Cumulative points per driver across seasons, with pit stop counts alongside.

#	Worksheet	Chart	Fields	Caption
5	GP Host Countries	Map	country	A world map of countries that have hosted Grand Prix rounds.
6	Driver Reliability (rename: Podiums by Driver)	Bar	driver, Podium	Podium finishes for each driver, showing who most consistently reaches the top three.
7	Driver Reliability (2) (rename: Podiums by Team)	Heat map	team, Podium	Podium finishes by team, showing which constructors dominate the podium.
8	Reliability Trend by Season (line)	Line	season, Reliability %	Share of entries that finished each season, showing whether reliability is improving.
9	Reliability Trend by Season (rename: Failure Type by Team and Circuit Type)	Table / bars	team, circuit_type, dnf_reason	DNF causes by team and circuit type, revealing where failures cluster.
10	Failure Type by Team and Circuit Type (rename: DNF by Driver and Race)	Table / bars	driver, race_id, dnf_reason	DNF occurrences by driver and race, filtered to retirements.

Delete the empty Sheet 12 before publishing.

2. Dashboards

All dashboards use a range size starting at 1300 x 860 pixels so they adapt to different screens.

Dashboard	Audience	Worksheets	Filters and actions
Dashboard 1	Team Performance Strategist	Team Championship Trajectory, Tyre Strategy, Talent Pools	Wet race filter; team selection
Dashboard 2	Broadcast Analyst	Driver Leaderboard, GP Host Countries, Podiums by Team	Use as filter on the leaderboard
Dashboard 3	Regulation Analyst	Podiums by Driver, Failure Type by Team and Circuit Type, Reliability Trend by Season	DNF reason filter
Dashboard 4	Reliability summary	Reliability Trend by Season	None

3. Story

A story presents the findings in order. Combine the two existing one-scene stories into a single story with these scenes:

Scene	Content	Caption to write
1	Introduction and overview	What the project covers and who it serves
2	Dashboard 1	Main finding for the strategist (team trajectory and tyre strategy)
3	Dashboard 2	Main finding for the broadcaster (leaderboard and podiums)
4	Dashboard 3	Main finding for the regulator (reliability and failure types)
5	Conclusion	Key takeaways

4. Design Guidelines Applied

- Consistent team colors across all worksheets.
- Short, descriptive titles and a caption under each chart.

- Filters grouped together at the side of each dashboard.
- Tooltips showing driver, team, season, and points.
- Layout containers so charts resize with the dashboard.

5. Performance Testing

Measure	Value
Amount of data rendered	6,464 race records, 17 seasons
Filters used	Wet race, DNF, DNF reason
Calculated fields	5 calculations plus 1 parameter-driven field
Worksheets	10
Dashboards	4
Stories	1 (after merging)

To keep the workbook fast, use an extract instead of a live connection, hide unused fields, and avoid more filters than needed.

6. Publishing

1. Choose File > Save to Tableau Public and sign in.
2. Open the published link and check that every dashboard and the story load.
3. Copy the public link: <PASTE-TABLEAU-PUBLIC-LINK-HERE>
4. Add screenshots of each dashboard to the GitHub repository.